

Croissant standard in the context of Dataverse, EOSC and beyond

Building global Linked Data landscape ready for ML

Slava Tykhonov / DANS-KNAW

Phil Durbin / Harvard IQSS

Croissant working group meeting

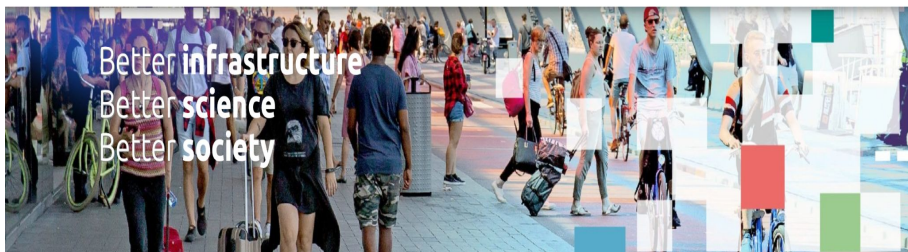
<https://mlcommons.org/croissant/>

20.03.2024

ODISSEI - Dutch national infrastructure for social sciences



RESEARCH ▾ FACILITIES ▾ ORGANISATION ▾ NEWS AND EVENTS ▾



ODISSEI

ODISSEI (Open Data Infrastructure for Social Science and Economic Innovations) is the national research infrastructure for the social sciences in the Netherlands. ODISSEI brings together researchers with the necessary data, expertise and resources to conduct ground-breaking research and embrace the computational turn in social enquiry.

Through ODISSEI, researchers have access to large-scale, longitudinal data collections as well as innovative and diverse new forms of data. These can be linked to administrative data at Statistics Netherlands (CBS). Combining data from a wide range of sources enables researchers to answer new, exciting, interdisciplinary research questions and to investigate existing questions in novel, new ways.



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Welcome to the prototype of the ODISSEI portal. The ODISSEI Portal combines metadata from a wide variety of research data repositories into a single interface, allowing for advanced semantic queries to support findability, and facilitate data access. The portal is under active development and features and content will be improved throughout the ODISSEI Roadmap project.



Search this dataverse... **Q** Advanced Search

☒ **Dataverses (19)**

☒ **Datasets (7,704)**

Dataverse Category
Organization or Institution (19)

Publication Year
2023 (259)
2022 (476)
2021 (546)
2020 (418)
2019 (343)

[More...](#)

Distributor Name
Centraal Bureau voor Statistiek (1,170)
CentERdata (277)
DataverseNL (184)
Stichting Nationaal Monument Kamp Amersfoort (73)
Centerdata (72)

1 to 10 of 7,723 Results

Sort ▾

Replication Data Netherlands-Iran Relations 1959-1979

Oct 10, 2023 - DANS



M. Warnaar, 2023, "Replication Data Netherlands-Iran Relations 1959-1979", <https://doi.org/10.17026/SS/M9AQZD>, ODISSEI Portal, V1

This dataset contains all unrestricted materials collected in the National Archives for the NWO Veni project The Familiar Other: Cultural Representations and Netherlands-Iran Relations, 1959-1979 which ran from 1 February 2017 to 5 July 2023 and resulted in the publication of the...

CBS (CBS)

Oct 5, 2023



This section of the portal includes metadata on data designs of microdata that can be available for researchers through the remote access facilities of CBS.

DANS (DANS)

Oct 5, 2023



This section of the portal includes metadata from the social science datasets available in the DANS Data Station Social Sciences and Humanities. More information is available at: <https://dans.knaw.nl/en/data-stations/social-sciences-and-humanities/>

HSN (HSN)

Oct 5, 2023

<https://portal.odissei-data.nl>

Croissant export in Dataverse

 **Dataverse**

Search ▾ User Guide Support Sign Up Log In

Cars

(Dataverse.org)

Root > Cars >

Cars

Version 1.0

 Admin, Dataverse, 2024, "Cars", <https://doi.org/10.5072/FK2/DZRHUP>, Root, V1, UNF:6:RPd9EWHSZwqUvRZuKTJmQg== [fileUNF]

[Cite Dataset ▾](#) [Learn about Data Citation Standards.](#)

[Access Dataset ▾](#)

Contact Owner Share

Dataset Metrics ⓘ

0 Downloads ⓘ

Description ⓘ Data about cars.

Subject ⓘ Other

License/Data Use Agreement  CC0 1.0

[Files](#) [Metadata](#) [Terms](#) [Versions](#)

Citation Metadata ▾

Persistent Identifier ⓘ	doi:10.5072/FK2/DZRHUP
Publication Date ⓘ	2024-03-14
Title ⓘ	Cars
Author ⓘ	Admin, Dataverse (Dataverse.org)
Point of Contact ⓘ	Use email button above to contact.
	Admin, Dataverse (Dataverse.org)
Description ⓘ	Data about cars.

[Export Metadata ▾](#)

- OAI_ORE
- DataCite
- OpenAIRE
- Schema.org JSON-LD
- DDI
- Dublin Core
- Croissant
- DDI HTML Codebook
- JSON

Mappings

Croissant
exporter

Code

```
@context:
  @language: "en"
  @vocab: "https://schema.org/"
  citeAs: "cr:citeAs"
  column: "cr:column"
  conformsTo: "dct:conformsTo"
  cr: "http://nlcommons.org/croissant/"
  data:
    @id: "cr:data"
    @type: "@json"
    dataBiases: "cr:dataBiases"
    dataCollection: "cr:dataCollection"
  dataType:
    @id: "cr:dataType"
    @type: "@vocab"
    dct: "http://purl.org/dc/terms/"
    extract: "cr:extract"
    field: "cr:field"
    fileProperty: "cr:fileProperty"
    fileObject: "cr:fileObject"
    fileSet: "cr:fileSet"
    format: "cr:format"
    includes: "cr:includes"
    isEnumeration: "cr:isEnumeration"
    jsonPath: "cr:jsonPath"
    key: "cr:key"
    md5: "cr:md5"
    parentField: "cr:parentField"
    path: "cr:path"
    personalSensitiveInformation: "cr:personalSensitiveInformation"
    recordSet: "cr:recordSet"
    references: "cr:references"
    regex: "cr:regex"
    repeated: "cr:repeated"
    replace: "cr:replace"
    sc: "https://schema.org/"
    separator: "cr:separator"
    source: "cr:source"
    subField: "cr:subField"
    transform: "cr:transform"
    wd: "https://www.wikidata.org/wiki/"
  @type: "sc:Dataset"
  conformsTo: "http://nlcommons.org/croissant/1.0"
  name: "Cars"
  version: "1.0"
  citeAs: "https://doi.org/10.5072/FK2/DZRHUP"
  distribution:
    0:
      @type: "cr:FileObject"
      @id: "stata13-auto.dta"
      name: "stata13-auto.dta"
      encodingFormat: "application/x-stata-13"
      md5: "7b1281ce6b469796837a835377338c5a"
      contentUrl: "stata13-auto.dta"
  recordSet:
    0:
      @type: "cr:RecordSet"
      field:
        0:
          @type: "cr:Field"
          name: "make"
          description: "Make and Model"
          dataType: "sc:Text"
          source:
            @id: "6"
            fileObject:
              @id: "stata13-auto.dta"
```

Experimental (graph) Croissant transformations in pyDataverse

```
Dataverse to Croissant.ipynb
File Edit View Insert Runtime Tools Help Cannot save changes

+ Code + Text Copy to Drive

[4] #!pip3 uninstall -y pyDataverse
#!pip3 install --upgrade --no-cache-dir git+https://github.com/Dans-labs/pyDataverse@croissant#egg=pyDataverse
from pyDataverse.Croissant import Croissant
import json

host = "https://dataverse.nl"
PID = "doi:10.34894/KMRAYH"
croissant = Croissant(host, PID)

#!pip3 install rdflib
from rdflib import Graph, plugin
from rdflib.serializer import Serializer
g = Graph().parse(data=croissant.get_record(), format='json-ld')

#print(g.serialize(format='json-ld', indent=4))
sparql_query = """
SELECT DISTINCT ?author
WHERE {
  ?node ?p ?label.
  ?node ?x ?author
  FILTER(STRSTARTS(str(?label), "https://schema.org/Person"))
}
"""
# Execute the SPARQL query and print results
results = g.query(sparql_query)

print("Searching for dataset authors in the graph:")
for row in results:
    print(row)

Searching for dataset authors in the graph:
(rdflib.term.URIRef('https://schema.org/Person'),)
(rdflib.term.Literal('Wonders, Ferry', lang='en'),)
(rdflib.term.Literal('Fontys Hogeschool', lang='en'),)
(rdflib.term.Literal('Raaij, Lars van', lang='en'),)
(rdflib.term.Literal('InnoBeweeLab', lang='en'),)
(rdflib.term.Literal('Murray Marshall, Teague', lang='en'),)
(rdflib.term.Literal('Gennip, Bart van', lang='en'),)
```

```
Dataverse to Croissant.ipynb
File Edit View Insert Runtime Tools Help Cannot save changes

+ Code + Text Copy to Drive

from google.colab import drive

#drive.mount('/content/drive')
print(json.dumps(croissant.get_record(), indent=4, default=str))

{
  "@type": "sc:Dataset",
  "name": "WijkBeweeKaart",
  "citeAs": "WijkBeweeKaart",
  "creator": [
    {
      "@type": "sc:Person",
      "name": "Wonders, Ferry",
      "affiliation": "Fontys Hogeschool"
    },
    {
      "@type": "sc:Person",
      "name": "Gennip, Bart van",
      "affiliation": "Fontys Hogeschool"
    },
    {
      "@type": "sc:Person",
      "name": "Murray Marshall, Teague",
      "affiliation": "Fontys Hogeschool"
    },
    {
      "@type": "sc:Person",
      "name": "Raaij, Lars van",
      "affiliation": "InnoBeweeLab"
    }
  ],
  "dateCreated": "2023-12-13 00:00:00",
  "datePublished": "2024-03-18 00:00:00",
  "description": "Hoe kan een digitale tool ontwikkeld worden om het bereik van het bewegonderzoek te vergroten onder",
  "isLiveDataset": true,
  "keywords": [
    "Lichaamsbeweging",
    "Stratum (Eindhoven)",
    "Wijkontwikkeling",
    "Demografisch onderzoek",
    "ICT-hulpmiddelen"
  ],
  "license": "http://creativecommons.org/licenses/by-nc-sa/4.0",
  "url": "https://doi.org/10.34894/KMRAYH",
}
```

FAIR Semantic Mappings

SEMAF: A Proposal for a Flexible Semantic Mapping Framework

Version: 1.0, March 2021

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This report was created under the responsibility of CLARIN ERIC (clarin@clarin.eu). For comments and questions please contact Daan Broeder d.g.broeder@uu.nl.

```
@prefix dct: <http://purl.org/dc/terms/> .
@prefix schema: <http://schema.org/> .
@prefix oa: <http://www.openarchives.org/ore/terms/> .
@prefix citation: <https://dataverse.org/schema/citation/> .
@prefix citationdesc: <https://dataverse.org/schema/citation/dsDescription#> .
@prefix ddialliance: <http://rdf-vocabulary.ddialliance.org/discovery#> .
@prefix odissi: <https://portal.odissi.nl/schema/dansRights#> .
@prefix mlcroissant: <https://mlcommons.org/croissant#> .
@prefix semaf: <http://semaf.com/> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .
```

```
# Semantic artifact mappings
# Name field in Croissant
mlcroissant:name a "name" ;
skos:exactMatch dct:title .

# Description field
mlcroissant:description a "description" ;
skos:exactMatch citation:dsDescriptionValue ;
skos:exactMatch schema:description ;
skos:exactMatch citationdesc:Text ;
skos:exactMatch citation:dsDescriptionText .

# URL field
mlcroissant:url a "url" ;
skos:exactMatch oa:describes .

# "citation": "https://dataverse.org/schema/citation/datasetContactName",
mlcroissant:citation a "citation" ;
skos:exactMatch citation:datasetContactName .

# "keywords": "https://dataverse.org/schema/citation/keywordValue",
mlcroissant:keywords a "keywords" ;
skos:exactMatch citation:keywordValue .
```

```
https://mlcommons.org/croissant#description http://www.w3.org/2004/02/skos/core#exactMatch https://dataverse.org/schema/citation/dsDescriptionValue
https://mlcommons.org/croissant#url http://www.w3.org/2004/02/skos/core#exactMatch http://www.openarchives.org/ore/terms/describes
https://mlcommons.org/croissant#description http://www.w3.org/2004/02/skos/core#exactMatch https://dataverse.org/schema/citation/dsDescriptionText
https://mlcommons.org/croissant#citation http://www.w3.org/1999/02/22-rdf-syntax-ns#type url
https://mlcommons.org/croissant#citation http://www.w3.org/1999/02/22-rdf-syntax-ns#type citation
https://mlcommons.org/croissant#citation http://www.w3.org/2004/02/skos/core#exactMatch https://dataverse.org/schema/citation/datasetContactName
https://mlcommons.org/croissant#name http://www.w3.org/2004/02/skos/core#exactMatch http://purl.org/dc/terms/title
https://mlcommons.org/croissant#name http://www.w3.org/1999/02/22-rdf-syntax-ns#type name
https://mlcommons.org/croissant#keywords http://www.w3.org/1999/02/22-rdf-syntax-ns#type keywords
https://mlcommons.org/croissant#description http://www.w3.org/1999/02/22-rdf-syntax-ns#type description
https://mlcommons.org/croissant#description http://www.w3.org/2004/02/skos/core#exactMatch https://dataverse.org/schema/citation/dsDescription#Text
https://mlcommons.org/croissant#description http://www.w3.org/2004/02/skos/core#exactMatch http://schema.org/description
https://mlcommons.org/croissant#keywords http://www.w3.org/2004/02/skos/core#exactMatch https://dataverse.org/schema/citation/keywordValue
```

```
s.get_fields(s.g, "https://mlcommons.org/croissant#keywords", SEARCH="SUBJECT", REPEATED=False)
```

```
['keywords', 'https://dataverse.org/schema/citation/keywordValue']
```

```
s.get_fields(s.g, "http://www.w3.org/1999/02/22-rdf-syntax-ns#type", SEARCH="PREDICATE", REPEATED=False)
```

```
['name', 'description', 'url', 'citation', 'keywords']
```



Croissant mappings in RDF are separated from the code and maintained on GitHub as a *semantic mapping resource*, incurring no maintenance costs for the software.

See reference:

Tykhonov, V. "CLARIAH integrations of Dataverse with SEMAF Semantic framework and SpaCy DANS Machine Learning library" <https://zenodo.org/records/8141719>

FAIR Signposting

FAIR Signposting Profile

Prepared by: Herbert Van de Sompel, Martin Klein, Shawn Jones, Michael L. Nelson, Simeon Warner, Anusuriya Devaraju, Robert Huber, Wilko Steinhoff, Vyacheslav Tykhonov, Luc Boruta, Enno Meijers, Stian Solland-Reyes, Mark Wilkinson

This version, created 20231002: <https://signposting.org/FAIR/>

[Previous version](#) created 20230519

[Version history](#)

Please provide feedback in the [GitHub Signposting repository](#)

This page details concrete recipes that platforms that host research outputs (e.g. data repositories, institutional repositories, publisher platforms, etc.) can follow to implement [Signposting](#), a lightweight yet powerful approach to increase the [FAIRness](#) of scholarly objects. Landing pages support humans that interact with scholarly objects on the web, providing descriptive metadata and links to content. These pages are not optimized for use by machine agents that navigate the scholarly web. For example, how can a robot determine which links on the myriad of landing pages lead to content and which to metadata? Signposting caters to machine agents by providing this information, and more, in a standards-based way. It contributes to FAIR's Findable, Accessible, and Reusable by uniformly conveying to machines what the persistent identifier of a scholarly object is, where its landing page is, where and what its content is, where metadata that describes it is, and what the persistent identifier of its author is. It conveys this by means of meaningful links that have web locations (HTTP URIs) as their target.



Image courtesy of Patrick Hochstetler.

As such it does significantly more than merely providing the information. It invites machine agents to follow the links to their target location on the web, and hopefully find further information and links there. It essentially provides them with a map to guide their travels across the scholarly web. Signposting contributes to FAIR's Interoperability through its uniform approach and because it is entirely based on widely implemented web protocols specified in IETF RFCs. As such, the interoperability that results from adopting it is not restricted to the scholarly landscape but encompasses the web at large.

The recipes provided in this document take a gradual approach towards increasing FAIRness for scholarly resources by means of Signposting:

- [Level 1](#) requires providing a minimal set of typed links via HTTP [Link](#) headers and/or HTML [<link>](#) elements. Despite its simplicity, it already significantly improves the ability of machine agents to navigate the scholarly web.
- [Level 2](#) requires providing a comprehensive set of typed links via a Link Set and making that Link Set discoverable. It increases the action radius of machine agents and provides them with a complete map of a scholarly object.

```
curl -I "https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/V3S58Q"
HTTP/2 200
date: Tue, 19 Mar 2024 21:33:43 GMT
content-type: text/html; charset=UTF-8
set-cookie: AWSALB=NUpmU5nriDhr0zipJ7uV4NVYLGJ0Gy+0fYVMbUC16H2QF4viy9NtXeQkBjPQA5htyPBjYrhLgL7WPkrEHQqr6Bk5isTnsKxgq73YDbP/wBXEHjRTeyUQmg/JKoLN; Expires=Tue, 26 Mar 2024 21:33:42 GMT; Path=/
set-cookie: AWSALBCORS=NUpmU5nriDhr0zipJ7uV4NVYLGJ0Gy+0fYVMbUC16H2QF4viy9NtXeQkBjPQA5htyPBjYrhLgL7WPkrEHQqr6Bk5isTnsKxgq73YDbP/wBXEHjRTeyUQmg/JKoLN; Expires=Tue, 26 Mar 2024 21:33:42 GMT; Path=/; SameSite=None; Secure
server: Apache
set-cookie: JSESSIONID=8a26916dfce9ceef6ba858487495; Path=/; Secure
link: <https://doi.org/10.7910/DVN/V3S58Q>;rel="cite-as", <https://dataverse.harvard.edu/api/access/datafile/8944173>;rel="item";type="application/json",<https://dataverse.harvard.edu/api/access/datafile/9651098>;rel="item";type="application/vnd.oasis.opendocument.spreadsheet", <https://doi.org/10.7910/DVN/V3S58Q>;rel="describedby";type="application/vnd.citationstyles.csl+json",<https://dataverse.harvard.edu/api/datasets/export?exporter=schema.org&persistentId=doi:10.7910/DVN/V3S58Q>;rel="describedby";type="application/ld+json", <https://schema.org/AboutPage>;rel="type",<https://schema.org/Dataset>;rel="type", <http://creativecommons.org/publicdomain/zero/1.0>;rel="license", <https://dataverse.harvard.edu/api/datasets/persistentId/versions/1.0/1inkset?persistentId=doi:10.7910/DVN/V3S58Q> ; rel="linkset";type="application/linkset+json"
```

<https://signposting.org/FAIR/>

Data Copilot: automated translation of vocabularies

Kenmerken, huur, kosten en weegfactor van een zelfstandige huurwoning in Nederland

Version 1.0



Centraal Bureau voor de Statistiek, 2022, "Kenmerken, huur, kosten en weegfactor van een zelfstandige huurwoning in Nederland", <https://doi.org/10.57934/0b01e41080745985>, ODISSEI Portal, V1

[Cite Dataset](#)

[Learn about Data Citation Standards.](#)

Contact Owner

Share

Description

Dit data ontwerp beschrijft een panel van zelfstandige huurwoningen van welke de huren en (service)kosten worden geregistreerd van het huidige verslagjaar en het vorige verslagjaar; Onder zelfstandige woning wordt een woning verstaan met eigen toegangsdeur, eigen toilet, eigen keuken en eigen badgelegenheid. De belangrijkste kenmerken van de woning zijn: bouwjaar, bouw type, aantal kamers, oppervlakte, puntenwaardering, energielabel, wel of niet geliberaliseerde woning en waarde van onroerende zaken (WOZ) waarde. Daarnaast wordt vastgelegd de gemeente waarin de woning zich bevindt en de eigenaar categorie om te bepalen of de woning een sociale of particuliere woning is.

Subject

Social Sciences

Keyword

Huurwoning, Huurstijging, Huurharmonisatie, Economie, bedrijven en nationale rekeningen, Consumentenprijzen, [Huurenquete](#)

License/Data Use Agreement

[CBS Licence](#)

[Access](#)

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[Export Metadata](#)

Citation Metadata

Persistent Identifier	doi:10.57934/0b01e41080745985
Publication Date	2022-07-01
Title	Kenmerken, huur, kosten en weegfactor van een zelfstandige huurwoning in Nederland
Alternative Title	HUURENQUETE

Curl

```
curl -X 'GET' \
'http://10.147.18.193:8008/translate?job=translate&customquery=Consumentenprijzen&language=en&format=json&context= Dit%20data%20ontwerp%20beschrijft' \
-H 'accept: application/json'
```

LLM API response:

```
{
  "English": "Consumer prices",
  "Dutch": "Consumentenprijzen",
  "German": "Verbraucherpreise",
  "French": "Prix des consommateurs",
  "Spanish": "Precios de los consumidores",
  "Ukrainian": "ціни споживання",
  "Russian": "Цены потребления",
  "Italian": "Prezzi dei consumatori",
  "Portuguese": "Preços dos consumidores",
  "Polish": "Ceny konsumpcyjne",
  "Czech": "Ceny spotřební",
  "Slovak": "Ceny spotreby",
  "Greek": "Τιμές των καταναλωτών",
  "Swedish": "Konsumentpriser",
  "Norwegian": "Forbrukspriser",
  "Danish": "Forbrugspriser",
  "Short description": "The prices that consumers pay for goods",
  "concept": "Consumentenprijzen",
  "md5": "64ee933b5e146cf713005cb8de3dc9be",
  "md5context": "1e4dd0b648a9a016f3a74d5c49ee0281",
  "lang": "nl"
}
```

Data Copilot: automated vocabularies enrichment

Original vocabulary



Enriched vocabulary

CBS classificaties en begrippen

Woordenlijsten over Feedback Help | in English auf Deutsch

Begrippen CBS

Content-taal Nederlands Zoek

Alfabetisch **Hierarchie** Nieuw

A B C D E F G H I J K L M N O
P Q R S T U V W Z

Aanhangwagen
Aanvullend pensioen
Aard van de collectie
Aardgas
Aardgascondensaat
Aardoliegrondstoffen
Aardolieproducten
ABR → Algemeen Bedrijven Register
ABW → Algemene bijstandswet
Academisch ziekenhuis
Achterstandsleerling
Activa
Acuut letsel
Additieven
Administratief personeel penitentiaire inrichting
Administratieve afvoering
Administratieve correctie
Administratieve opname
Adoptie
Adoptie via de rechtbank
ADV → Arbeidsduurverkortung
ADV-dagen
Adviesprijs
Afdeling bestuursrechtspraak van de Raad van State →
Raad van State, afdeling bestuursrechtspraak
Afdoeing door de rechter
Afdoeing door het Openbaar Ministerie

Industrie en energie > Aardgascondensaat

VOORKEURSTERM **Aardgascondensaat**

TYPE CBS begrip

DEFINITIE Lichte koolwaterstoffen die als bijproduct vrijkomen bij de winning van aardgas of aardolie. Het zijn onder andere ethaan, propaan, butaan en pentaan.

BREDER CONCEPT Industrie en energie

BEREIKSNOTITIE Aardgascondensaat dat bij normale temperatuur en druk gasvormig is, wordt voor transport vloeibaar gemaakt.

BRON <https://www.cbs.nl/nl-nl/onze-diensten/methoden/begrippen/aardgascondensaat>

IDENTIFIER f4670a56-fd4d-4d21-81d6-4c81a9f45d82-nl-nl

VERSIE 1

IN ANDERE TALEN **Natural gas liquids** Engels

URI <https://vocab.cbs.nl/def/begrippen/1847>

DOWNLOAD DIT CONCEPT: RDF/XML Turtle JSON-LD Gemaakt 23-03-2016, laatst gewijzigd 23-02-2022

Skosmos

Vocabularies About Feedback Help | Interface language: English

CBS Begrippen 2

Content language Dutch Zoek

Alfabetical **Hierarchy**

A B C D E F G H I J K L M N O
P Q R S T U V W Z

Aanhangwagen
Aanvullend pensioen
Aard van de collectie
Aardgas
Aardgascondensaat
Aardoliegrondstoffen
Aardolieproducten
ABR → Algemeen Bedrijven Register
ABW → Algemene bijstandswet
Academisch ziekenhuis
Achterstandsleerling
Activa
Acuut letsel
Additieven
Administratief personeel penitentiaire inrichting
Administratieve afvoering
Administratieve correctie
Administratieve opname
Adoptie
Adoptie via de rechtbank
ADV → Arbeidsduurverkortung
ADV-dagen
Adviesprijs
Afdeling bestuursrechtspraak van de Raad van State →
Raad van State, afdeling bestuursrechtspraak
Afdoeing door de rechter
Afdoeing door het Openbaar Ministerie
Afdragen btw

Industrie en energie > Aardgascondensaat

PREFERRED TERM **Aardgascondensaat**

TYPE CBS begrip

DEFINITION Lichte koolwaterstoffen die als bijproduct vrijkomen bij de winning van aardgas of aardolie. Het zijn onder andere ethaan, propaan, butaan en pentaan.

BROADER CONCEPT Industrie en energie

SCOPE NOTE Aardgascondensaat dat bij normale temperatuur en druk gasvormig is, wordt voor transport vloeibaar gemaakt.

SOURCE <https://www.cbs.nl/nl-nl/onze-diensten/methoden/begrippen/aardgascondensaat>

IDENTIFIER f4670a56-fd4d-4d21-81d6-4c81a9f45d82-nl-nl

VERSION 1

IN OTHER LANGUAGES **天然气沉淀** Chinese
Natural gas liquids English
Condensat de gaz naturel French
Erdgasdampf German

URI <https://vocab.cbs.nl/def/begrippen/1847>

We use LLM to translate Dataverse keywords and link to SKOSMOS concepts in the specific “context”

Data Copilot: AI powered taxonomy creator

Input: Wikipedia or any page



Output: ontology “from scratch”

WIKIPEDIA The Free Encyclopedia

Search Wikipedia Search

COVID-19

190 languages

Contents hide

- (Top)
- Nomenclature
- Symptoms and signs
- Cause
- Pathophysiology
- Diagnosis
- Prevention
- Treatment
- Prognosis and risk factors
- Mortality
- History
- Misinformation
- Other species
- Research
- See also
- References
- Further reading
- External links

From Wikipedia, the free encyclopedia

For the pandemic caused by the said disease, see *COVID-19 pandemic*. For other diseases caused by coronaviruses, see *Coronavirus diseases*.

Coronavirus disease 2019 (COVID-19) is a **contagious disease** caused by the virus **SARS-CoV-2**. The first known case was **identified in Wuhan, China**, in December 2019.^[8] The disease quickly spread worldwide, resulting in the **COVID-19 pandemic**.

The **symptoms of COVID-19** are variable but often include fever,^[7] cough, headache,^[8] fatigue, **breathing difficulties**, loss of smell, and loss of taste.^{[9][10][11]} Symptoms may begin one to fourteen days after exposure to the virus. At least a third of people who are infected **do not develop noticeable symptoms**.^[12] Of those who develop symptoms noticeable enough to be classified as patients, most (81%) develop mild to moderate symptoms (up to mild **pneumonia**), while 14% develop severe symptoms (**dyspnea**, **hypoxia**, or more than 50% lung involvement on imaging), and 5% develop critical symptoms (**respiratory failure**, **shock**, or **multiorgan dysfunction**).^[14] Older people are at a higher risk of developing severe symptoms. Some people continue to experience a range of effects (**long COVID**) for months or years after infection, and damage to organs has been observed.^[15] Multi-year studies are underway to further investigate the long-term effects of the disease.^[16]

COVID-19 transmits when infectious particles are breathed in or come into contact with the eyes, nose, or mouth. The risk is highest when people are in close proximity, but small **airborne** particles containing the virus can remain suspended in the air and travel over longer distances, particularly indoors. Transmission can also occur when people touch their eyes, nose or mouth after touching surfaces or objects that have been contaminated by the virus. People remain contagious for up to 20 days and can spread the virus even if they do not develop symptoms.^[17]

Testing methods for COVID-19 to detect the virus's nucleic acid include **real-time reverse transcription polymerase chain reaction (RT-PCR)**,^{[18][19]} **transcription-mediated amplification**,^{[18][9][20]} and reverse transcription loop-mediated isothermal amplification (RT-LAMP)^{[18][16]} from a nasopharyngeal swab.^[21]

Coronavirus disease 2019 (COVID-19)

Other names COVID, (the) coronavirus

Transmission and life-cycle of SARS-CoV-2, which causes COVID-19

Pronunciation /kəˈroʊnəveɪrəs/ /koʊvɪdˈnæɪtən, ˈkɔvɪd-^[7]

Specialty Infectious disease

Symptoms Fever, cough, fatigue, shortness of breath, vomiting, loss of taste or smell; some cases asymptomatic^{[2][9]}

Complications Pneumonia, sepsis, ARDS, kidney failure, respiratory failure, pulmonary fibrosis, CKS, MIS-C, long COVID

Usual onset 2–14 days (typically 5) after infection

COVID-19 thesaurus

Content language English

Search

Alphabetical Hierarchy

A B C D E F G H I K L M N O P Q R S T U V W Y 0-9

A close match
A group of people
ACE-2 upregulation
ACE2 receptor
ACE2 receptors
active substance
actual mortality rate
Acute cardiac injury
Acute encephalopathy
Acute infection
Acute myocardial injuries
acute respiratory disease
Acute respiratory illness
Adaptation
Additional control measure
Adenovirus
Adults
Advanced age
Advanced age (above 65 years of age)
Advice
advised to
aerosol droplet
age
Age
Age structure
Agenda 2030
Aging
Agriculture

PREFERRED TERM

actual mortality rate

SCOPE NOTE

The actual mortality rate refers to the number of deaths per unit of time (usually per 100,000 people) in a given population, taking into account all deaths, regardless of whether they were reported or not. This is in

IDENTIFIER

DID

IN OTHER LANGUAGES

skutečná úmrtnost	Czech
virkelig dødelighed	Danish
werkelijke sterftecijfer	Dutch
taux de mortalité réel	French
tatsächliche Sterberate	German
σταθερή θανάτωση	Greek
tasso di mortalità reale	Italian
sann dødelighet	Norwegian
rzeczywista śmiertelność	Polish
taxa de mortalidade real	Portuguese
реальный уровень смертности	Russian
skutočná úmrtnosť	Slovak
tasa de mortalidad real	Spanish
verkelig dödlighet	Swedish
справжня смертність	Ukrainian

URI

<https://id.now.museum/cf17d5abb5a7de9703de9b15c8274d05>

LLM workflow to convert Web pages into SKOS vocabularies linked with Croissant fields

Dataverse integration with Skosmos is term lookup based

Subject * ?

Keyword ?

Vocabulary ?

Term ?



Vocabulary URL ?

Vocabulary ?

Term ?



Vocabulary URL ?

- COVID-19
- COVID-19 pandemic
- COVID-19 pandemic in India
- Covi
- Covi
- Covi
- SARS-CoV-2



Resulting in TermURI



Related Publication ?

Citation ?



Semantic Search for the Dutch data landscape

ODISSEI portal updated with Multilingual search engine: <https://dans.knaw.nl/en/news/improved-search-and-more-data-to-find/>

In addition to social sciences metadata from CBS, LISS, DataverseNL and DANS the metadata from the Historical Sample of the Netherlands (HSN) is now included. The Portal also harvests the newly released **DANS Data Station Social Sciences and Humanities** which has replaced the DANS archive EASY.

Improved search for CBS metadata

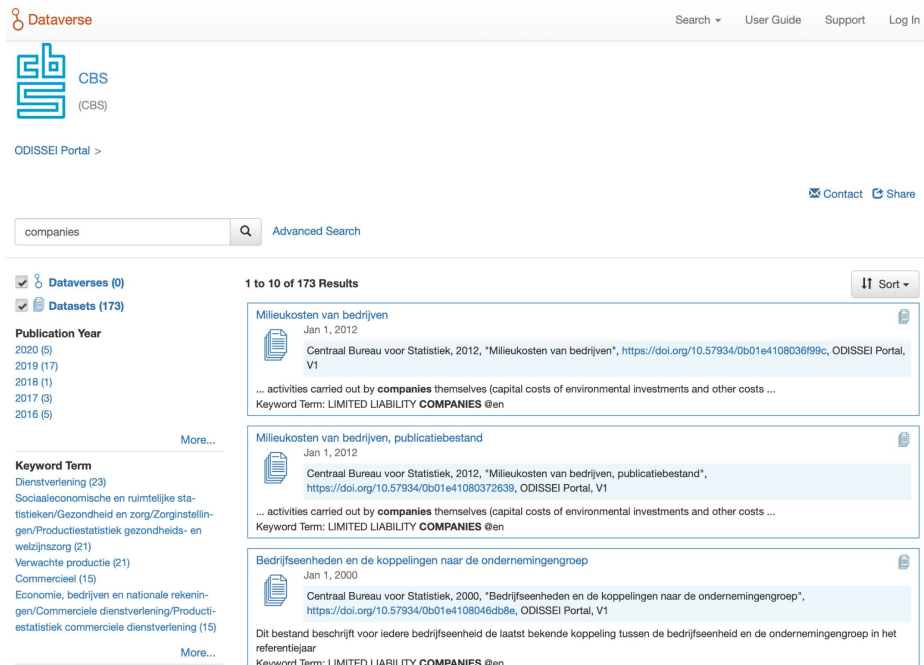
In this updated version of the ODISSEI Portal prototype, improvements are made to better find metadata related to CBS microdata sets. There is much more information included on the variables which are now visible in the metadata block "Variable information". It is now also possible to search for metadata records based on the short names used in the CBS Remote Access environment (e.g. KINDOUDERTAB).

Multilingual search for CBS keywords

A great improvement is that the Portal now supports multilingual search for CBS keywords. The CBS metadata is enriched by mapping the available Dutch keywords to keywords in the European Social Sciences Language Thesaurus (ELSST). In ELSST translations are available into other languages and this allows you to find Dutch metadata records when searching with English terms. For example, when you are in the CBS part of the ODISSEI Portal, you can search for "employment" and find datasets related to this, even though the metadata is only available in Dutch. This feature will be extended and improved in the next few months.

Further developments

Until the end of 2024, the Portal will be further developed and improved by ODISSEI, DANS, VU and SURF as part of the ODISSEI Roadmap project. The collected metadata will be enriched with existing controlled vocabularies to increase the findability of the information. More metadata providers will be added to the Portal with the ultimate goal of giving researchers access to information about all relevant social science datasets in the Netherlands. Data providers from the ODISSEI community are encouraged to get in touch with the Portal team to make their metadata available in the Portal. Last but not least, the functionalities around the data access broker will be developed, which should enable users to request data directly from the various data providers.



The screenshot shows the ODISSEI Portal search results page. At the top, there is a navigation bar with the Dataverse logo, a search bar, and links for User Guide, Support, and Log In. Below the navigation bar, the CBS logo is displayed, along with the text "ODISSEI Portal >". The search results are displayed in a table with columns for Publication Year, Keyword Term, and a list of results. The search term "companies" is entered in the search bar. The results are sorted by relevance, showing 1 to 10 of 173 results. The first result is "Milieukosten van bedrijven" (Environmental costs of companies) from 2012, published by the Centraal Bureau voor de Statistiek. The second result is "Milieukosten van bedrijven, publicatiebestand" (Environmental costs of companies, publication file) from 2012, also published by the Centraal Bureau voor de Statistiek. The third result is "Bedrijfseenheden en de koppelingen naar de ondernemingsgroep" (Business units and the connections to the business group) from 2000, published by the Centraal Bureau voor de Statistiek. Each result includes a brief description and a link to the full record.

Data Copilot: AI powered knowledge graph “miner”

Input: any sentence from text



Output: concepts with their relationships

[illegible]

LLM is building graph with “facts” using relationships from known vocabularies. Does the “fact statement” belong to Croissant’ Responsible AI?

DDI in Dataverse - breaking the bias

(variables, categories, units, ...)



Export Metadata ▼

OAI_ORE

DataCite

OpenAIRE

Schema.org JSON-LD

DDI

Dublin Core

DDI HTML Codebook

JSON

```
<var ID="v63828" name="gender" intrvl="contin">
  <location fileid="f181579"/>
  <category>
    <catValu>1</catValu>
    <labl level="category">male, or</labl>
  </category>
  <catValu>2</catValu>
  <labl level="category">female</labl>
  </category>
  <catValu>3</catValu>
  <labl level="category">other</labl>
  </category>
  <varFormat type="numeric"/>
  <notes subject="Universal Numeric Fingerprint" level="variable" type="Dataverse:UNF13:MDMaBjEgQoqHvUjPISkWa==</notes>
</var>
<var ID="v63998" name="marit_a" intrvl="contin">
  <location fileid="f181579"/>
  <category>
    <catValu>1</catValu>
    <labl level="category">never married</labl>
  </category>
  <catValu>2</catValu>
  <labl level="category">married</labl>
  </category>
  <catValu>3</catValu>
  <labl level="category">divorced</labl>
  </category>
  <catValu>4</catValu>
  <labl level="category">widowed</labl>
  </category>
  <varFormat type="numeric"/>
  <notes subject="Universal Numeric Fingerprint" level="variable" type="Dataverse:UNF13:MDMaBjEgQoqHvUjPISkWa==</notes>
</var>
```

Dataverse-Data-Curation-Tool

The Data Curation Tool (DCT) allows data owners and curators to view summary statistics for variables and to create and edit variable-level metadata for any tabular file in a data set. This stand-alone component is built to complement [The Dataverse Project](#). The Data Curation tool is integrated into dataverse for .tab files under the configure button.

The DCT is an Angular application and uses the Angular Material Design component library.

Responsible AI in Croissant?

Search

Items Per Page: 5 1 - 7 of 21

Edit Variable

☐	ID	Name	Label	Weight	
☐	v457580	RID	RpsRespondent	No Weight	
☐	v457594	Q1	Have you, or has anyone in your household had a fever, that is, a temperature above 38 degrees Celsius or about 100 degrees Fahrenheit, in the past week?	No Weight	
☐	v457599	Q2	Are you, or is anyone in your household currently suffering from a new cough in the past week?	No Weight	
☐	v457596	Q3	Are you, or is anyone in your household currently suffering from new headaches in the past week?	No Weight	
☐	v457593	Q4	Are you, or is anyone in your household suffering from a new sore throat in the past week?	No Weight	
☐	v457587	Q5	Are you, or is anyone in your household suffering from a loss of taste or smell in the past week?	No Weight	
☐	v457595	Q6	Are you, or is anyone in your household suffering from new diarrhea in the past week?	No Weight	

ID: v457599
Name: Q2

Label
Are you, or is anyone in your household currently suffering from a new cough in the past week?

Literal Question

Interviewer Instructions

Post Question

Universe

Notes
UNF:6:M6R5ZcCIBShqNvwelw==

Group(s)
Add to Group

No Weight Assigned Is Weight

Croissant and DDI

(Data Documentation Initiative)

Responsible AI

“As AI advances at rapid speed there is increased recognition among researchers, practitioners and policy makers that we need to explore, understand, manage, and assess [its economic, social, and environmental impacts](#). One of the main instruments to operationalise responsible AI (RAI) is dataset documentation.

This is how Croissant helps address RAI:

1. *It proposes a machine-readable way to capture and publish metadata about ML datasets – this makes existing documentation solutions like [Data Cards](#) easier to publish, share, discover, and reuse;*
2. *It records at a granular level how a dataset was created, processed and enriched throughout its lifecycle – this process is meant to be automated as much as possible by integrating Croissant with popular ML frameworks. By allowing the metadata to be loaded automatically, Croissant also enables developers to compute RAI metrics automatically and systematically, identifying potential data quality issues to be fixed.*

Croissant is designed to be modular and extensible. One such extension is the Croissant RAI vocabulary, which addresses 7 specific use cases, starting with the data life cycle, data labeling, and participatory scenarios to AI safety and fairness evaluation, traceability, regulatory compliance and inclusion. More details are available in the . We welcome additional extensions from the community to meet the needs of specific data modalities (e.g. audio or video) and domains (e.g. geospatial, life sciences, cultural heritage).”

Croissant spec v1.0



CESSDA DATA CATALOGUE (CDC) DDI3.3 PROFILE

DDI_XPath	Required	UI Label	Type	Repeatable	Usage note
/ddi:DDIInstance/@xml:lang	Recommended	Study description available in	Attribute		ISO 639-1 codes are strongly encouraged to be used
/ddi:DDIInstance/@xsi:schemaLocation	Recommended		Attribute		System-required information - nearly always "ddi:instance-3.3 http://www.ddialliance.org/Specification/DDI-Lifecycle/3.3/XMLSchema/instance.xsd"
/ddi:DDIInstance/r:Citation/r:Title/r:String	Optional		Content element	No	Title of the CDC XML document - note that the Study Title goes in /ddi:DDIInstance/s:StudyUnit/r:Citation/r:Title/r:String
/ddi:DDIInstance/r:Citation/r:Title/r:String/@xml:lang	Mandatory if 'r:Title/r:String' element is present		Attribute		ISO 639-1 codes are strongly encouraged to be used
/ddi:DDIInstance/r:ResourcePackage/pi:PhysicalInstance/r:Citation/r:Language	Recommended	Language of data file(s)	Content element	Yes	ISO 639-1 codes are strongly encouraged to be used if available for the language.
/ddi:DDIInstance/s:StudyUnit/r:UserID	Mandatory	Study number / PID	Content element	Yes	This element serves two use cases. (1) Unique archival number. If no such number available, a PID may be used alternatively (and in such a case, the PID must be captured in this element "as well as" in "/ddi:DDIInstance/s:StudyUnit/r:Citation/r:InternationalIdentifier/r:IdentifierContent"). The @typeOfUserID attribute must be "StudyNumber". (2) A URL or URN (a reference to a web resource that specifies its location) linking to the study description on the SP website. CDC provides a link from the study information there to the study description on the SP website, to allow users to access the data. [This element not visible in the UI, it is only needed for the link]. The @typeOfUserID attribute must be "URLServiceProvider".
/ddi:DDIInstance/s:StudyUnit/r:UserID/@typeOfUserID	Mandatory		Attribute		Must be specified when the "ddi:DDIInstance/s:StudyUnit/r:UserID" element is used for the unique archival number.
/ddi:DDIInstance/s:StudyUnit/r:UserID/@typeOfUserID	Mandatory		Attribute		Must be specified when the "ddi:DDIInstance/s:StudyUnit/r:UserID" element is used for the URL of the study description at the SP website.
/ddi:DDIInstance/s:StudyUnit/r:Citation/r:Title/r:String	Mandatory	Study title	Content element	Yes	Title of the Study (as opposed to the title of the XML document).
/ddi:DDIInstance/s:StudyUnit/r:Citation/r:Title/r:String/@xml:lang	Mandatory		Attribute		Language of the study title. ISO 639-1 codes are strongly encouraged to be used.
/ddi:DDIInstance/s:StudyUnit/r:Citation/r:Creator/r:CreatorReference	Recommended	Creator	Container element	Yes	Reference to Principal investigator Person OR Institution
/ddi:DDIInstance/s:StudyUnit/r:Citation/r:Creator/r:CreatorReference/r:TypeOfObject	Mandatory of 'r:CreatorReference' is present.	Creator	Content element	Yes	Type of object being referenced - in this case, it should always be either "Individual" or "Organization".
/ddi:DDIInstance/s:StudyUnit/r:Citation/r:Publisher/r:PublisherReference	Recommended	Publisher	Container element	Yes	A reference to the name of the institution publishing the metadata, i.e. the name of the actual CESSDA Service Provider providing the metadata information. Only on study level.
/ddi:DDIInstance/s:StudyUnit/r:Citation/r:Publisher/r:PublisherReference/r:TypeOfObject	Mandatory if 'r:PublisherReference' is present	Publisher	Content element	Yes	Type of object being referenced - in this case, it should always be either "Individual" or "Organization".

[CESSDA DDI profile](#)

DDI-CDI and Croissant

DDI – Cross Domain Integration: Introduction

Contents

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III. Purpose	2
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C. Provenance and Process Description.....	3
D. Foundational Metadata	3
E. Interoperability, Sustainability, and Alignment with Other Standards.....	4
V. DDI - CDI and the Suite of DDI Specifications	4

I. Overview

The DDI – Cross Domain Integration (DDI – CDI) specification provides a model for working with a wide variety of research data across many scientific and policy domains. It provides a level of detail which supports machine-actionable processing of data, both within and between systems, and is designed to be easily aligned with other standards.

It focuses on the key elements of the data management challenges facing research today: an exact understanding of data in a wide variety of formats, coming from many different sources. Two elements are critical for dealing with these challenges: a flexible means of describing data that can reveal the connections between the same data existing in different formats, and a means of describing the provenance of the data at a detailed (but comprehensible) level: the processes which produced it must be made transparent.

DDI - CDI covers these areas in a fashion intended to make it optimally useful to modern systems, which often employ a variety of models, and comply with a range of related specifications for both functions related to data description and process/provenance. The model is designed to be easy to fit into such systems, by aligning with relevant external standards, and to be alignable with them into the future.



DDI – CDI: Integrating Data for Better Science

variableMeasured	
@type	PropertyValue
name	eventCode
description	A code identifying the type of event that has occurred
unitText	discrete
additionalType	https://ddicore.org/dataStructureComponent/measureComponent?obo:entity=transaction#obo:entityCharacteristic=classification
propertyID	https://hdss.org/eventType
identifier	
@type	PropertyValue
url	https://hdss.org/eventType/enumeration/ENU
description	Starting event for all individuals present at the baseline census of the surveillance area. It corresponds to the date on which the individual was first observed to be present in the surveillance area during the baseline census.
identifier	
@type	PropertyValue
url	https://hdss.org/eventType/birth/BTH
description	The birth of an individual to a resident female.
identifier	
@type	PropertyValue
url	https://hdss.org/eventType/inmigration/IMG
description	The event of migrating into the surveillance area.
identifier	
@type	PropertyValue
url	https://hdss.org/eventType/outmigration/OMG
description	The event of migrating out of the surveillance Destination (same as for Orain) area.

Use case: AI Assistant for datasets

Example: generate SQL statement to get how many Titanic passengers survived from the second class

Prompt:

```
titanic_answer:
{
  action: 'text-generation'
  instruction: 'Given the following SQL table, your job is to write queries given a user's request. CREATE TABLE {} ({} ) \n'
  template: 'Write a SQL query that returns - {}'
  query: 'How many people survived from the second class?'
  device_map: 'auto'
}
```

Request:

```
curl http://0.0.0.0:8008/transformers?job=titanic_answer
```

<|system|>

Given the following SQL table, your job is to write queries given a user's request. CREATE TABLE df (PassengerId BIGINT, Pclass BIGINT, Name VARCHAR, Sex VARCHAR, Age DOUBLE, SibSp BIGINT, Parch BIGINT, Ticket VARCHAR, Fare DOUBLE, Cabin VARCHAR, Embarked VARCHAR)

<|user|>

Write a SQL query that returns - How many people survived from the second class?

<|assistant|>

To answer this question, we need to filter the data to only include passengers from the second class (Pclass = 2) and then count the number of passengers who survived (Survived = 1).

Here's the SQL query:

```
SELECT COUNT(*) FROM df WHERE Pclass = 2 AND Survived = 1;
```

This query will return the total number of passengers who survived from the second class.

Use case: list of questions your data can answer

<|system|>

Given the following SQL table, your job is to write queries given a user's request. CREATE TABLE df (PassengerId BIGINT, Pclass BIGINT, Name VARCHAR, Sex VARCHAR, Age DOUBLE, SibSp BIGINT, Parch BIGINT, Ticket VARCHAR, Fare DOUBLE, Cabin VARCHAR, Embarked VARCHAR)

<|user|>

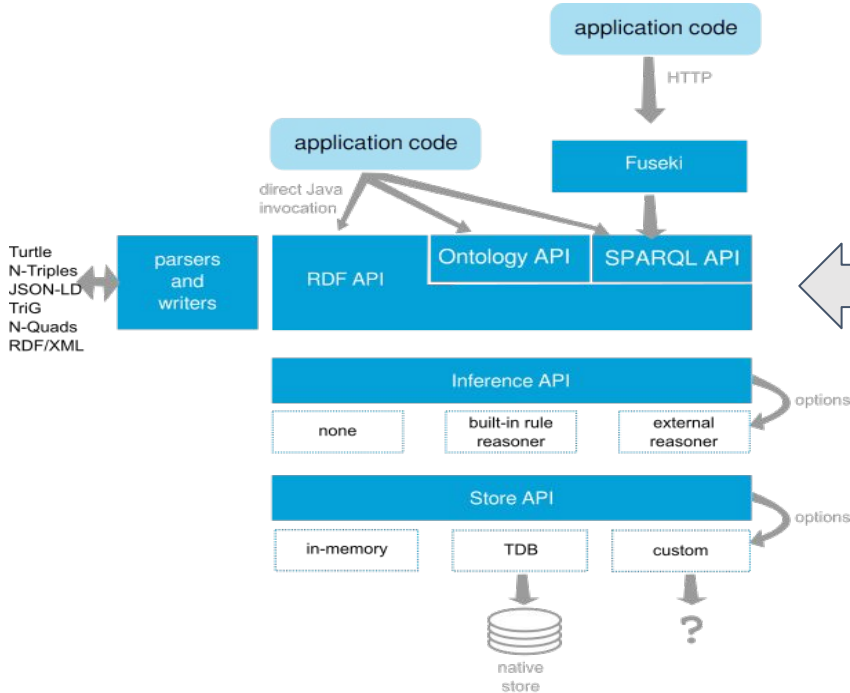
Write a SQL query that returns - Give me back possible questions to be answered with table df

<|assistant|>

I do not have the context of your specific use case or requirements. however, based on the given table, here are some possible questions that can be answered using sql queries:

1. what is the average fare for passengers in the first class (pclass = 1)?
2. how many female passengers (sex = 'female') were traveling with at least one sibling (sibsp > 0)?
3. what is the most common embarkation port for passengers in the third class (pclass = 3)?
4. how many passengers embarked from southampton (embarked ='s') and what is the average fare for them?
5. what is the highest fare paid by a passenger in the second class (pclass = 2)?
6. how many passengers were traveling alone (sibsp = parch = 0)?
7. what is the total number of passengers who embarked from quebec (embarked = 'q')?
8. what is the average age for passengers in the second class (pclass = 2) who embarked from southampton (embarked ='s')?

Where do we go from here? LLM+KG integration



SPARQL query

To try out some SPARQL queries against the selected dataset, enter your query here.

EXAMPLE QUERIES
[Selection of triples](#) [Selection of classes](#)

PREFIXES
[rdf](#) [rdfs](#) [owl](#) [xsd](#) [⊞](#)

SPARQL ENDPOINT

CONTENT TYPE (SELECT)

CONTENT TYPE (GRAPH)

```
5 PREFIX citation: <https://dataverse.org/schema/citation/>
6 PREFIX ore: <http://www.openarchives.org/ore/1.0/terms/>
7 PREFIX distributor: <https://dataverse.org/schema/citation/distributor#>
8 PREFIX variable: <https://portal.staging.odissel.nl/schema/variableInformation/variable#>
9 SELECT ?id ?title ?distributorname
10 WHERE {
11   ?id dct:terms:title ?title;
12       dct:terms:subject ?subject;
13       schema:datePublished ?date;
14       citation:Distributor ?distributor.
15   ?distributor distributor:Name ?distributorname.
16   ?id schema:datePublished "2010-01-01"
17 }
```

QUERY RESULTS
[Table](#) [Raw Response](#) [Download](#)

Showing 1 to 1 of 1 entries

id	title	distributorname
<doi:10.57934/Os01e41080216906>	"Minimumloonklasse van banen van werknemers"	"Centraal Bureau voor Statistiek"

Search: Show 50 entries

Meet our “built-in” Dataverse triple store: Jena Fuseki

Example: AI generated Graph on Data Management topic

```
ns:00e6c77594614c63f1a543c7056159f9 skos:prefLabel "Information Retrieval" ;
  skos:related ns:091fa9121c047db1dd48c3e2ab5f3c91,
    "Machine Learning" .

ns:019efa159436b35b7e2d3f9dbe4117d0 ns1:Better "Data Organization" ;
  ns1:Enhances "Research Productivity" ;
  ns1:Increases "Research Efficiency" ;
  ns1:Used_for "Knowledge Organization" ;
  skos:prefLabel "Data Annotation" ;
  skos:related ns:99f8bf10da893ec5c236164ed14f22b4,
    "Research Trends" .

ns:022cf5438481b74f225bb36577f41699 skos:prefLabel "data archiving" ;
  skos:related ns:81513effdf5790b79549208838404407,
    ns:d06d61becdaf07c6424d0c4a03e929c2,
    "loss",
    "preservation" .

ns:03c8731cc88a2a4b0479a12f1f831df6 skos:narrower ns:245089aea5f60411b7dca63c73d0293f,
  "Research Notes" ;
  skos:prefLabel "Lab Notebooks" .

ns:0b259705b755c4cbd672e6663ae31905 ns1:Is_Part_Of "Backup" ;
  skos:prefLabel "Version Control" .

ns:0bb6b854f8c6f925221ed67e4ecc75fd skos:prefLabel "Expert Systems" ;
  skos:related ns:0a40e3c91a3a55c9a37428c6d194d0e5,
    "AI" .
```

Next to SKOS and OWL new
types of “hidden” relationships
detected in data by AI!

Is Croissant X.X good place to
populate AI generated
metadata?

More missing features in Croissant implemented in Dataverse

Sensitive vs Restricted files

Embargo info

Provenance, data ownership transfer

Primary and secondary (derivative) datasets

Questions?